

Explainable AutoML

This survey is part of a master's thesis investigating how different levels of explainability influence the perception of machine learning models in a financial context. You will be presented with several model explanations and asked to evaluate them based on different criteria such as interpretability, plausibility, trustworthiness, and usefulness. The survey includes examples from two application domains: credit risk assessment and marketing prediction. Prior domain knowledge is not required. Please evaluate each explanation from your professional perspective and based on the information provided. There are no right or wrong answers. We are interested in your personal assessment and experience.

Please review the notebooks and try to understand how the models work. The outputs generated by each notebook are part of the explanation and are intended to help you understand the model and its predictions. Each level folder contains one notebook. Start with **Level 0 (Baseline: Logistic Regression)**, review the notebook and its outputs, and then answer the corresponding questionnaire questions. Afterward, continue with **Level 1**, **Level 2**, **Level 3**, and **Level 4** in order. Please note that the models in **Levels 1–3** are based on the same underlying model. Each level only adds additional explainability features. **Level 4** uses a restricted model search space, resulting in a different model selection process and explainability setup. The objective is to evaluate how understandable and useful the different levels of explainability are from your perspective.

IMPORTANT: Please focus on understanding how the model arrives at its predictions rather than evaluating how accurate or effective the model is. The goal is to assess whether you can understand the reasoning and information provided to explain how the prediction was generated.

All responses will be collected anonymously and used exclusively for research purposes. The survey will take approximately 45 minutes to complete. Thank you for your participation.

* Erforderlich

Personal data

1. Your role *

- ☐ Risk Management
- ☐ Data Science / Analytics
- ☐ Sonstiges

2. Work Experience *

- ☐ < 2 years
- ☐ 2–5 years
- ☐ 6–10 years
- ☐ > 10 years

3. Familiarity

	Not at all familiar			Moderately familiar			Extremely familiar
Familiarity with Machine Learning	☐	☐	☐	☐	☐	☐	☐
Familiarity with Credit Risk Models	☐	☐	☐	☐	☐	☐	☐
Familiarity with Python	☐	☐	☐	☐	☐	☐	☐

Use Case 1: PD

You will be presented with several model outputs related to the prediction of the **Probability of Default (PD)**. The Probability of Default represents the estimated likelihood that a customer will default on a loan within the next 12 months. Such models are widely used in credit risk management and are subject to regulatory requirements. The models shown in this study differ not only in their predictive algorithms but also in the amount and type of explanations provided to the user.

What you will see: Each use case contains information about a customer and the model's predicted default risk. Depending on the explainability level, additional information may be provided. Examples include: Calibration Machine learning models often produce probability estimates that are not perfectly aligned with observed outcomes. Therefore, calibration techniques are applied to ensure that predicted probabilities better reflect real-world default rates. For example, if a group of customers receives a predicted PD of 5%, approximately 5% of these customers should actually default over time. Margin of Conservatism (MoC) Banking regulations require institutions to account for potential uncertainty in model estimates. Therefore, additional conservative adjustments may be applied to the predicted PD. Two forms are used in this study:

- **MoC A:** Adjustment for identified methodological weaknesses or model limitations.
- **MoC C:** Adjustment for uncertainty caused by limited data, sampling effects, or estimation errors.

These adjustments increase the final PD to ensure a more conservative risk assessment. PD Floor Regulatory frameworks often require a minimum default probability, even for very low-risk customers. A PD floor prevents the model from producing unrealistically low risk estimates and ensures a minimum level of conservatism.

Risk Grades: Some model versions transform continuous PD values into discrete risk classes or grades. Customers with similar risk profiles are grouped into the same grade, making model outputs easier to interpret compare.

Feature Importance: Certain models provide information about which customer characteristics had the strongest influence on the prediction.

Local Explanations: Some models provide explanations for individual customer prediction. These explanations indicate which factors increased or decreased the predicted default risk for the specific customer being analyzed. Methods such as SHAP and LIME may be used to generate these explanations.

Explainability Levels:

Level 0 – Baseline Model. **Model:** Logistic Regression The participant receives only the basic model output and standard model information without additional explainability techniques. Also, the preprocessing "WoE" for LogReg can be found in this folder. Level 1 – **Model:** H2O AutoML with built-in explainability features Additional information such as feature importance and model-specific interpretation tools are provided.

Level 2 – **Model:** Level 1 + SHAP and LIME In addition to the inherent explanations, post-hoc explanation methods are used to explain individual predictions and model behavior.

Level 3 – **Model:** Level 2 + Business-Oriented Explanations. Additional risk-management-oriented information provided, including:

- Scorecards & Risk grades & PD floor adjustments
- Calibration analyses
- Margin of Conservatism adjustments (MoC A / MoC C)
- Business-oriented visualizations/reports

Level 4 – Restricted Search Space AutoML. The AutoML search process is restricted to inherently interpretable model families (e.g., linear models and tree-based models).

Level 0

4. Block A: Interpretability *

	Not at all			Neutral		Totally
I understand how the model arrived at its prediction	☐	☐	☐	☐	☐	☐
The explanation clearly identifies the most important risk factors	☐	☐	☐	☐	☐	☐
I could explain this prediction to a colleague	☐	☐	☐	☐	☐	☐
I could justify this prediction in a professional discussion	☐	☐	☐	☐	☐	☐

5. Block B: Regulatory Plausibility *

	Not at all			Neutral		Totally
The explanation provides sufficient justification for the prediction	☐	☐	☐	☐	☐	☐
The prediction appears traceable and auditable	☐	☐	☐	☐	☐	☐
The explanation would support model validation activities	☐	☐	☐	☐	☐	☐
The explanation would be suitable for use in a regulated banking environment	☐	☐	☐	☐	☐	☐

6. Block C: Trust *

	Not at all			Neutral		Totally
I trust this prediction	☐	☐	☐	☐	☐	☐
The model behaves in a reliable manner	☐	☐	☐	☐	☐	☐
I feel confident using the model output	☐	☐	☐	☐	☐	☐

7. Block D: Acceptance *

	Not at all			Neutral		Totally
I would use this model output in practice	☐	☐	☐	☐	☐	☐
I would recommend the use of this model	☐	☐	☐	☐	☐	☐

Level 1

8. Block A: Interpretability *

	Not at all			Neutral		Totally
I understand how the model arrived at its prediction	☐	☐	☐	☐	☐	☐
The explanation clearly identifies the most important risk factors	☐	☐	☐	☐	☐	☐
I could explain this prediction to a colleague	☐	☐	☐	☐	☐	☐
I could justify this prediction in a professional discussion	☐	☐	☐	☐	☐	☐

9. Block B: Regulatory Plausibility *

	Not at all			Neutral		Totally
The explanation provides sufficient justification for the prediction	☐	☐	☐	☐	☐	☐
The prediction appears traceable and auditable	☐	☐	☐	☐	☐	☐
The explanation would support model validation activities	☐	☐	☐	☐	☐	☐
The explanation would be suitable for use in a regulated banking environment	☐	☐	☐	☐	☐	☐

10. Block C: Trust *

	Not at all			Neutral		Totally
I trust this prediction	☐	☐	☐	☐	☐	☐
The model behaves in a reliable manner	☐	☐	☐	☐	☐	☐
I feel confident using the model output	☐	☐	☐	☐	☐	☐

11. Block D: Acceptance *

	Not at all			Neutral		Totally
I would use this model output in practice	☐	☐	☐	☐	☐	☐
I would recommend the use of this model	☐	☐	☐	☐	☐	☐

Level 2

12. Block A: Interpretability *

	Not at all			Neutral		Totally
I understand how the model arrived at its prediction	☐	☐	☐	☐	☐	☐
The explanation clearly identifies the most important risk factors	☐	☐	☐	☐	☐	☐
I could explain this prediction to a colleague	☐	☐	☐	☐	☐	☐
I could justify this prediction in a professional discussion	☐	☐	☐	☐	☐	☐

13. Block B: Regulatory Plausibility *

	Not at all			Neutral		Totally
The explanation provides sufficient justification for the prediction	☐	☐	☐	☐	☐	☐
The prediction appears traceable and auditable	☐	☐	☐	☐	☐	☐
The explanation would support model validation activities	☐	☐	☐	☐	☐	☐
The explanation would be suitable for use in a regulated banking environment	☐	☐	☐	☐	☐	☐

14. Block C: Trust *

	Not at all			Neutral		Totally
I trust this prediction	☐	☐	☐	☐	☐	☐
The model behaves in a reliable manner	☐	☐	☐	☐	☐	☐
I feel confident using the model output	☐	☐	☐	☐	☐	☐

15. Block D: Acceptance *

	Not at all			Neutral		Totally
I would use this model output in practice	☐	☐	☐	☐	☐	☐
I would recommend the use of this model	☐	☐	☐	☐	☐	☐

Level 3

16. Block A: Interpretability *

	Not at all			Neutral		Totally
I understand how the model arrived at its prediction	☐	☐	☐	☐	☐	☐
The explanation clearly identifies the most important risk factors	☐	☐	☐	☐	☐	☐
I could explain this prediction to a colleague	☐	☐	☐	☐	☐	☐
I could justify this prediction in a professional discussion	☐	☐	☐	☐	☐	☐

17. Block B: Regulatory Plausibility *

	Not at all			Neutral		Totally
The explanation provides sufficient justification for the prediction	☐	☐	☐	☐	☐	☐
The prediction appears traceable and auditable	☐	☐	☐	☐	☐	☐
The explanation would support model validation activities	☐	☐	☐	☐	☐	☐
The explanation would be suitable for use in a regulated banking environment	☐	☐	☐	☐	☐	☐

18. Block C: Trust *

	Not at all			Neutral		Totally
I trust this prediction	☐	☐	☐	☐	☐	☐
The model behaves in a reliable manner	☐	☐	☐	☐	☐	☐
I feel confident using the model output	☐	☐	☐	☐	☐	☐

19. Block D: Acceptance *

	Not at all			Neutral		Totally
I would use this model output in practice	☐	☐	☐	☐	☐	☐
I would recommend the use of this model	☐	☐	☐	☐	☐	☐

Level 4

20. Block A: Interpretability *

	Not at all			Neutral		Totally
I understand how the model arrived at its prediction	☐	☐	☐	☐	☐	☐
The explanation clearly identifies the most important risk factors	☐	☐	☐	☐	☐	☐
I could explain this prediction to a colleague	☐	☐	☐	☐	☐	☐
I could justify this prediction in a professional discussion	☐	☐	☐	☐	☐	☐

21. Block B: Regulatory Plausibility *

	Not at all			Neutral		Totally
The explanation provides sufficient justification for the prediction	☐	☐	☐	☐	☐	☐
The prediction appears traceable and auditable	☐	☐	☐	☐	☐	☐
The explanation would support model validation activities	☐	☐	☐	☐	☐	☐
The explanation would be suitable for use in a regulated banking environment	☐	☐	☐	☐	☐	☐

22. Block C: Trust *

	Not at all			Neutral		Totally
I trust this prediction	☐	☐	☐	☐	☐	☐
The model behaves in a reliable manner	☐	☐	☐	☐	☐	☐
I feel confident using the model output	☐	☐	☐	☐	☐	☐

23. Block D: Acceptance *

	Not at all			Neutral		Totally
I would use this model output in practice	☐	☐	☐	☐	☐	☐
I would recommend the use of this model	☐	☐	☐	☐	☐	☐

Use Case2: Marketing

You will be presented with several model outputs related to a **direct marketing campaign**. The objective of the model is to predict whether a customer is likely to respond positively to a marketing offer. Such models are commonly used to improve campaign efficiency by identifying customers with the highest probability of conversion. The models shown in this study differ not only in their predictive algorithms but also in the amount and type of explanations provided to the user. What you will see Depending on the explainability level, you may encounter:

- **Response Probability:** The estimated likelihood that a customer will respond positively to the campaign.
- **Calibration:** Additional adjustments that improve the reliability of predicted probabilities, ensuring that predicted response rates better match observed outcomes.
- **Marketing Grades:** Customers grouped into risk or opportunity segments based on their predicted response probability.
- **Budget-Based Customer Examples:** Example customers located at different campaign budget thresholds (e.g., highly attractive targets, borderline cases, or low-priority customers).
- **Feature Importance:** Information about which customer characteristics most influenced the prediction.
- **Local Explanations:** Individual explanations generated using methods such as SHAP or LIME, showing why a specific customer received a particular prediction.
- **Campaign Performance Metrics:** Business-oriented evaluation tools such as Lift Charts, Gains Charts, and customer ranking analyses that illustrate how effectively the model identifies likely responders.

Explainability Levels: Level 0 – Baseline **Model:** Logistic Regression Basic prediction output without additional explainability methods. In this folder, also the preprocessing for the LogReg can be found. Level 1 – Inherent Explainability **Model:** H2O AutoML Built-in explainability features such as feature importance and model-specific interpretation tools. Level 2 – Post-hoc Explainability **Model:** Level 1 + SHAP and LIME Additional local and global explanations of model behavior and individual predictions. Level 3 – Business Explainability **Model:** Level 2 + Business-Oriented Explanations Additional information such as:

- Marketing grades
- Lift charts
- Gains charts
- Budget-based customer examples
- Customer segmentation analyses
- Business-oriented performance reports

Level 4 – Restricted Search Space. **Model:** Level 3 + Restricted AutoML Search Space The AutoML process is restricted to inherently interpretable model families (e.g., linear and tree-based models).

Level 0 -Marketing

24. Block A: Interpretability *

	Not at all			Neutral		Totally
I understand how the model arrived at its prediction	☐	☐	☐	☐	☐	☐
The explanation clearly identifies the most important risk factors	☐	☐	☐	☐	☐	☐
I could explain this prediction to a colleague	☐	☐	☐	☐	☐	☐
I could justify this prediction in a professional discussion	☐	☐	☐	☐	☐	☐

25. Block B: Regulatory Plausibility *

	Not at all			Neutral		Totally
The explanation provides sufficient justification for the prediction	☐	☐	☐	☐	☐	☐
The prediction appears traceable and auditable	☐	☐	☐	☐	☐	☐
The explanation would support model validation activities	☐	☐	☐	☐	☐	☐
The explanation would be suitable for use in a regulated banking environment	☐	☐	☐	☐	☐	☐

26. Block C: Trust *

	Not at all			Neutral		Totally
I trust this prediction	☐	☐	☐	☐	☐	☐
The model behaves in a reliable manner	☐	☐	☐	☐	☐	☐
I feel confident using the model output	☐	☐	☐	☐	☐	☐

27. Block D: Acceptance *

	Not at all			Neutral		Totally
I would use this model output in practice	☐	☐	☐	☐	☐	☐
I would recommend the use of this model	☐	☐	☐	☐	☐	☐

Level 1- Marketing

28. Block A: Interpretability *

	Not at all			Neutral		Totally
I understand how the model arrived at its prediction	☐	☐	☐	☐	☐	☐
The explanation clearly identifies the most important factors	☐	☐	☐	☐	☐	☐
I could explain this prediction to a colleague	☐	☐	☐	☐	☐	☐
I could justify this prediction in a professional discussion	☐	☐	☐	☐	☐	☐

29. Block B: Regulatory Plausibility *

	Not at all			Neutral		Totally
The explanation provides sufficient justification for the prediction	☐	☐	☐	☐	☐	☐
The prediction appears traceable and auditable	☐	☐	☐	☐	☐	☐
The explanation would support model validation activities	☐	☐	☐	☐	☐	☐
The explanation would be suitable for use in a regulated banking environment	☐	☐	☐	☐	☐	☐

30. Block C: Trust *

	Not at all			Neutral		Totally
I trust this prediction	☐	☐	☐	☐	☐	☐
The model behaves in a reliable manner	☐	☐	☐	☐	☐	☐
I feel confident using the model output	☐	☐	☐	☐	☐	☐

31. Block D: Acceptance *

	Not at all			Neutral		Totally
I would use this model output in practice	☐	☐	☐	☐	☐	☐
I would recommend the use of this model	☐	☐	☐	☐	☐	☐

Level 2 - Marketing

32. Block A: Interpretability *

	Not at all			Neutral		Totally
I understand how the model arrived at its prediction	☐	☐	☐	☐	☐	☐
The explanation clearly identifies the most important factors	☐	☐	☐	☐	☐	☐
I could explain this prediction to a colleague	☐	☐	☐	☐	☐	☐
I could justify this prediction in a professional discussion	☐	☐	☐	☐	☐	☐

33. Block B: Regulatory Plausibility *

	Not at all			Neutral		Totally
The explanation provides sufficient justification for the prediction	☐	☐	☐	☐	☐	☐
The prediction appears traceable and auditable	☐	☐	☐	☐	☐	☐
The explanation would support model validation activities	☐	☐	☐	☐	☐	☐
The explanation would be suitable for use in a regulated banking environment	☐	☐	☐	☐	☐	☐

34. Block C: Trust *

	Not at all			Neutral		Totally
I trust this prediction	☐	☐	☐	☐	☐	☐
The model behaves in a reliable manner	☐	☐	☐	☐	☐	☐
I feel confident using the model output	☐	☐	☐	☐	☐	☐

35. Block D: Acceptance *

	Not at all			Neutral		Totally
I would use this model output in practice	☐	☐	☐	☐	☐	☐
I would recommend the use of this model	☐	☐	☐	☐	☐	☐

Level 3 - Marketing

36. Block A: Interpretability *

	Not at all			Neutral		Totally
I understand how the model arrived at its prediction	☐	☐	☐	☐	☐	☐
The explanation clearly identifies the most important factors	☐	☐	☐	☐	☐	☐
I could explain this prediction to a colleague	☐	☐	☐	☐	☐	☐
I could justify this prediction in a professional discussion	☐	☐	☐	☐	☐	☐

37. Block B: Regulatory Plausibility *

	Not at all			Neutral		Totally
The explanation provides sufficient justification for the prediction	☐	☐	☐	☐	☐	☐
The prediction appears traceable and auditable	☐	☐	☐	☐	☐	☐
The explanation would support model validation activities	☐	☐	☐	☐	☐	☐
The explanation would be suitable for use in a regulated banking environment	☐	☐	☐	☐	☐	☐

38. Block C: Trust *

	Not at all			Neutral		Totally
I trust this prediction	☐	☐	☐	☐	☐	☐
The model behaves in a reliable manner	☐	☐	☐	☐	☐	☐
I feel confident using the model output	☐	☐	☐	☐	☐	☐

39. Block D: Acceptance *

	Not at all			Neutral		Totally
I would recommend the use of this model	☐	☐	☐	☐	☐	☐
I would use this model output in practice	☐	☐	☐	☐	☐	☐

Level 4 - Marketing

40. Block A: Interpretability *

	Not at all			Neutral		Totally
I understand how the model arrived at its prediction	☐	☐	☐	☐	☐	☐
The explanation clearly identifies the most important factors	☐	☐	☐	☐	☐	☐
I could explain this prediction to a colleague	☐	☐	☐	☐	☐	☐
I could justify this prediction in a professional discussion	☐	☐	☐	☐	☐	☐

41. Block B: Regulatory Plausibility *

	Not at all			Neutral		Totally
The explanation provides sufficient justification for the prediction	☐	☐	☐	☐	☐	☐
The prediction appears traceable and auditable	☐	☐	☐	☐	☐	☐
The explanation would support model validation activities	☐	☐	☐	☐	☐	☐
The explanation would be suitable for use in a regulated banking environment	☐	☐	☐	☐	☐	☐

42. Block C: Trust *

	Not at all			Neutral		Totally
I trust this prediction	☐	☐	☐	☐	☐	☐
The model behaves in a reliable manner	☐	☐	☐	☐	☐	☐
I feel confident using the model output	☐	☐	☐	☐	☐	☐

43. Block D: Acceptance *

	Not at all			Neutral		Totally
I would use this model output in practice	☐	☐	☐	☐	☐	☐
I would recommend the use of this model	☐	☐	☐	☐	☐	☐

Overall questions

The following questions no longer refer to any specific models.

Final questions

44. In which use case was explainability more important?

- ☐ PD
- ☐ Marketing
- ☐ Equally important

45. Which explainability level would you recommend for PD?

- ☐ Level 0
- ☐ Level 1
- ☐ Level 2
- ☐ Level 3
- ☐ Level 4

46. Which explainability level would you recommend for Marketing

- ☐ Level 0
- ☐ Level 1
- ☐ Level 2
- ☐ Level 3
- ☐ Level 4

47. Did the benchmark model (Level 0) influence your assessment?

- ☐ Yes
- ☐ Partially
- ☐ No

Open questions

48. What aspects of the explanation were most useful?

49. What information was missing or insufficient for assessing the prediction?

50. At what point does explainability become excessive?

Dieser Inhalt wurde von Microsoft weder erstellt noch gebilligt. Die von Ihnen übermittelten Daten werden an den Formulareigentümer gesendet.

